Mohammed Yahya Alharbi

Data Engineer

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SUMMARY

Junior Data Engineer focused on building reliable, end-to-end pipelines that turn raw data into analytics-ready models. Delivered cloud lakehouse and ELT projects with incremental loading, historical tracking, and automated data quality checks, backed by database administration aligned with Saudi data governance standards (NDMO/NDI).

TECHNICAL SKILLS

Programming: Python, SQL, Jinja, YAML
Databases: Microsoft SQL Server, PostgreSQL, MySQL, Azure SQL Database
Database Administration: Backup & Recovery, User & Role Management, Database Security, Performance Monitoring
Data Governance: NDMO Data Management Standards, National Data Index (NDI) – Data Storage Domain, Compliance
Azure: Azure Data Factory, ADLS Gen2, Azure SQL Database, Logic Apps
Data Platforms: Databricks (Unity Catalog, Autoloader, Lakeflow, Asset Bundles), Snowflake, Microsoft Fabric
Processing: PySpark, Spark Structured Streaming
Transformation & Orchestration: dbt Core, Apache Airflow
Data Modeling: Medallion Architecture, Star Schema, SCD Type 1/2, One Big Table (OBT)
AWS: S3, Glue, Athena, Lambda, IAM, EC2
DevOps: Git, GitHub, Azure DevOps, Linux, CI/CD
Soft Skills: Continuous Learning, Attention to Detail, Problem-Solving Under Pressure, Clear Communication, Social Intelligence

PROJECTS

WaveLake: Music Streaming Lakehouse | Azure & Databricks
Tech Stack: *Azure Data Factory, ADLS Gen2, Azure SQL Database, Databricks, Unity Catalog, PySpark, Lakeflow, Logic Apps, GitHub*

- Built an end-to-end lakehouse pipeline (Medallion Architecture) from Azure SQL Database to ADLS Gen2 and Databricks.
- Created parameterized ADF pipelines for incremental multi-table loading, with Logic Apps alerts and GitHub version control.
- Processed data with PySpark and Autoloader under Unity Catalog, using Jinja templates for reusable SQL.
- Built a Star Schema in the Gold layer with SCD Type 2 and data quality checks using Lakeflow Declarative Pipelines.

StayScope: Airbnb Analytics Platform | dbt, Snowflake & AWS S3
Tech Stack: *dbt Core, Snowflake, AWS S3, SQL, Jinja, Python (uv), Git*

- Loaded Airbnb listings, hosts, and bookings from AWS S3 into Snowflake and modeled them end to end with dbt Core.
- Designed bronze, silver, and gold layers with incremental models, merges on business keys, and custom schema routing.
- Generated a One Big Table and fact model from config-driven Jinja loops, with reusable macros and ephemeral models.
- Tracked dimension history with dbt snapshots (SCD Type 2) and added singular data quality tests on source data.

FlightPath: Airline Data Platform | Azure Data Factory & Azure DevOps
Tech Stack: *Azure Data Factory, ADLS Gen2, Azure SQL Database, Mapping Data Flows, Delta Lake, Logic Apps, Azure DevOps*

- Ingested airline, flight, and booking data from on-prem files (self-hosted IR), a REST API, and Azure SQL into ADLS Gen2.
- Built dynamic, parameterized pipelines with ForEach and If Condition, plus incremental loads tracked in a JSON file in the lake.
- Refined data with Mapping Data Flows into a Delta Lake silver layer using upserts, then aggregated it into gold tables.
- Orchestrated a parent pipeline with triggers and Logic Apps alerts; versioned in Azure DevOps with PRs and ARM templates.

EXPERIENCE

Technical Camp Instructor (Part-Time) | Tuwaiq Academy, Qassim **Jul 2026 – Aug 2026**

- Designed and delivered a 4-week curriculum (Cybersecurity, AI, Arduino, Game Development) for 15 students.
- Guided students through hands-on final projects; received a certificate of appreciation from Tuwaiq Academy.

Summer Intern | Your House Smart, Saudi Arabia **Jun 2025 – Jul 2025**

- Compiled and organized a database of 100 potential partners using advanced Excel functions.
- Managed e-commerce platform operations to improve store organization and user experience.

EDUCATION

Bachelor of Information Technology | Qassim University, Saudi Arabia **Jan 2026**
GPA: 4.29/5.0 | Very Good with Second Class Honors

Data Engineering Bootcamp | Saudi Digital Academy (SDA) – WeCloudData **Aug 2026 – Oct 2026**
Gained skills in Data Ingestion, ETL/ELT, Data Transformation, Data Modeling, and Data Quality with Python, Azure, Snowflake, and dbt.

CERTIFICATIONS & TRAINING

- Data Engineering Program | Athar Academy
- Data Engineering Bootcamp | Saudi Digital Academy (SDA)
- Comprehensive Database Management Framework – National Data Index (NDI) | Bayan Academy
- AWS Certified Cloud Practitioner | Saudi Digital Academy
- DevOps Fundamentals in Cloud Environments | MCIT
- SQL Database Application using Python | Tuwaiq Academy

LANGUAGES

Arabic (Native) | English (Professional Working Proficiency)